ProjectBayMax · build notebook

Day 10 · Joint and payload hazards

R0.3 · 2026-11-02 · Owner: Faisal Farouq

Day 10 result: **PASS** (desk work: hazard list and control ideas). Severity, likelihood and residual risk stay **UNASSESSED** for a qualified reviewer.

Full table: `day_010_motion_payload_hazards.csv` (15 hazards, MP-01 to MP-15). On Day 13 they get permanent IDs (JB-H-21 onward).

The short version

- **The drives are strong enough to crush a hand.** A knee at its rated torque pushes about 760 N at 10 cm from the joint, and a hip-roll joint about 1,140 N. Force limits can't make that safe, so the controls are **guards, gaps and distance**.
- **A runaway joint is fast.** At no-load speed a hip-roll joint turns 7 rad/s, so the leg tip could move about 6 m/s. That's why speed limits, a watchdog and an independent e-stop come before any motion, and why first tests use a restraint.
- **The handling stand is low-energy but still pinches.** The object weighs 19.6 N, but the actuator's push force can be hundreds of newtons, so the gap and the actuator choice matter more than the object.
- **People will want to touch it.** The pinned arms and tray look like handles. Briefings, barrier signs and the keep-out zone deal with that.

The 15 hazards

ID	Where	Hazard	Key number	Control idea	Check
MP-01	Knee	Finger or hand in the knee-pad gap	760 N rated, 2,280 N peak at 0.10 m	Guard or bellows; else gap ≥ 25 mm (fingers)	Days 97, 317
MP-02	Ankle	Finger between shin and foot cover	400 N rated, 1,200 N peak at 0.10 m	Same as knee	Days 95, 97, 317
MP-03	Hip	Hand between thighs or thigh and pelvis	1,140 N rated, 3,420 N peak at 0.10 m	Guard; else gap ≥ 100 mm (hand)	Days 93, 97, 317
MP-04	Foot	Person's foot stepped on	up to 539.55 N through one foot	Keep-out zone, barrier	Days 136, 261
MP-05	Swinging leg	Leg strikes a person	≈0.45 m/s planned foot speed	Keep-out zone; speed limits	Days 131, 136
MP-06	Any joint	Runaway joint	20.9 / 10.5 / 7.0 rad/s no-load; leg tip ≈6 m/s	Firmware + file speed limits, watchdog, independent e-stop, end stops, restraint	Days 131, 221, 233
MP-07	Shoulder pin	Pinch when posing; arm drops if pin lost	arm ≈1 kg (assumed)	Captive pin with clip; preflight check	Days 87, 261

ID	Where	Hazard	Key number	Control idea	Check
MP-08	Shell seams	Panels collide, crack or pinch skin	—	Swept-clearance study; rounded edges	Days 97, 309, 317
MP-09	Cables	Cable snag cuts a sensor or power line	detect sensor loss ≤ 100 ms	Service loops; bend check; stale-sensor detection	Days 97, 216
MP-10	Tray	Dropped object	21.6 J to the floor from 1.1 m; 2.9 J onto a catch tray 0.15 m below	Lip + pin; catch tray; unbreakable object	Days 201, 298
MP-11	Handling actuator	Hand pinched by the tray	push force = datasheet (often 100s of N) vs 24.5 N held	Lowest push force that still holds ≥ 98 N; guard or ≥ 100 mm gap; limit switches	Days 187, 201, 299
MP-12	Stand	Stand tips over	7.35 N m; 10 kg stand needs base half-width > 0.075 m	Wide base or bolted	Day 201
MP-13	Handling power loss	Tray drops when power cut	24.5 N held; ≥ 98 N holding screen	Holding force or brake; power-off test	Days 296, 299
MP-14	Arms, tray, stand	Used as a handhold	not designed to hold a person	Briefing, sign, no contact	Days 139, 333
MP-15	Mode	Walking with the object or docked to the stand	—	Payload and support interlocks block walking	Days 237, 299

Gap values come from ISO 13854 (minimum gaps to avoid crushing parts of the body: fingers 25 mm, hand 100 mm). Treat them as starting values; the mechanical reviewer confirms them.

How the numbers were worked out

- **Pinch force** = joint torque ÷ distance from the joint axis. Example: 76 N m ÷ 0.10 m = 760 N. Rated values from Day 6: 40 N m direct, 76 N m at 2:1, 114 N m at 3:1. Peak values: 120, 228, 342 N m.
- **Runaway speed**: RobStride 04 no-load 200 rpm = 20.9 rad/s; ÷2 = 10.5; ÷3 = 7.0 rad/s. Leg tip at 0.86 m: 7.0 × 0.86 ≈ 6.0 m/s.
- **Planned swing**: 0.144 m in 0.6 s with a smooth profile peaks at 1.875 × the average speed: 1.875 × 0.24 = 0.45 m/s.
- **Drop energy** = m g h: 2 × 9.81 × 1.1 = 21.6 J; 2 × 9.81 × 0.15 = 2.9 J.

How this links to the existing register

Existing hazard	New detail added
JB-H-02 pinch / shell	MP-01, 02, 03, 07, 08, 11
JB-H-03 runaway	MP-06, 09
JB-H-05 payload / misuse	MP-10 to 15
JB-H-17 power loss	MP-13

Existing hazard	New detail added
JB-H-20 people get close	MP-04, 05, 14

Outside help

A mechanical reviewer confirms the gap sizes and guard designs (Days 97 and 199) and the handling holding force (Day 296). A controls or safety reviewer confirms the runaway protections (Day 129). No one is named yet (D-02).

Checks

Check	Result
Every hazard has a control idea, a test ID and a reviewer role	PASS
No gap size, force limit or residual risk presented as approved	PASS
Person lifting and walking with cargo listed as excluded uses	PASS (MP-14, MP-15)

Sources: R0.1 hazard register JB-H-02, 03, 05, 17, 20; R0.1 verification register; Day 6 R0.3 drive data; Day 9 keep-out sketch; ISO 13854 (minimum gaps to avoid crushing; values to be confirmed by the reviewer).